



Phase modulators in hybrid silicon and lithium niobate thin films

QINGYUN LI,¹ HOUBIN ZHU,¹ HONGHU ZHANG,¹ LUTONG CAI,^{2,3} 
AND HUI HU^{1,*} 

¹School of Physics, State Key Laboratory of Crystal Materials, Shandong University, No. 27 South Shanda Road, Jinan 250100, China

²Key Laboratory of Opto-Electronic Information Technology of Ministry of Education, School of Precision Instruments and Opto-electronics Engineering, Tianjin University, 92 Weijin Road, Tianjin 300072, China

³Key Laboratory of Integrated Opto-Electronic Technologies and Devices in Tianjin, School of Precision Instruments and Opto-electronics Engineering, Tianjin University, 92 Weijin Road, Tianjin 300072, China

*hhu@sdu.edu.cn

Abstract: The hybridization of mono-crystalline silicon and lithium niobate thin films (Si-LNOI) can combine the physical properties of Si and the optical properties of LN, and it has emerged as a new material platform for integrated photonics. In this paper, phase modulators in Si-LNOI were demonstrated. First, the phase modulator was designed. According to the simulation, the Si loading strip waveguide had a small mode area due to the large refractive index of Si. This allowed a small electrode gap that resulted in enhancement of the overlap of the optical field and the electrostatic field, and the $V_{\pi}L$ of the phase modulator could be as small as 3.3 V·cm. Second, phase modulators with a Si loading strip waveguide with a top width of 0.5 μm were fabricated by plasma etching, and the $V_{\pi}L$ of the phase modulator was measured to be 4.74 V·cm. This study provides useful information for the devices in the Si-LNOI.

© 2022 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

1. Introduction

Lithium niobate (LN) crystal has excellent electro-optic, acousto-optic, and nonlinear optical properties, and it is widely used in the field of integrated photonics [1–2]. However, the low refractive index contrast of conventional optical waveguides based on LN bulk material exhibits low confinement for optical modes [3–6], which greatly limits the miniaturization of waveguide devices and application in integrated photonic devices. The lithium niobate thin films (lithium niobate on insulator, LNOI) retain the physical properties of LN bulk materials and have large refractive index contrast [7–8]. With the industrialization of LNOI and the breakthrough of micromachining technology, many high-performance photonic devices based on LNOI have been reported, e.g., electro-optic modulators [9–11], micro-ring and micro-disk resonators [12–14], and nonlinear conversion devices [15–19].

SOI (silicon on insulator) has been widely used in integrated circuits and integrated photonics due to its excellent electrical properties and mature processing technology. However, Si lacks $\chi^{(2)}$ nonlinear optical effects, which hinders its application in photonics. Numerous efforts were made to add $\chi^{(2)}$ nonlinear optical effects to SOI, and several hybrid waveguide devices using LN on SOI (LN-SOI) were introduced. The linearity of a ring modulator in LN-SOI was over an order of magnitude greater than that in SOI using the plasma dispersion effect [20]. The Mach-Zehnder (M-Z) modulator in LN-SOI achieves a $V_{\pi}L$ of 6.7 V·cm and modulation bandwidth of 100 GHz [21]. Direct bonding generally requires that the two bonded surfaces be flat. Devices in LN-SOI usually bonded LN thin films on the top of patterned SOI, which makes the direct bonding process challenging. Furthermore, the fabrication and connection of electrodes would be another challenge. Therefore, a universal material platform formed by the combination of LN and Si is highly preferred. Hybrid mono-crystalline silicon and

lithium niobate thin films (Si-LNOI) have been reported recently, that can combine the physical properties of Si and the optical properties of LN and provide more flexibility in device design [22]. Compared to amorphous Si (a-Si) deposited on LNOI [23], mono-crystalline Si thin films have lower transmission loss. The Si-LNOI material platform could be compatible with silicon photonics and attractive for fabrication of future photonic devices. A few reports on photonic devices fabricated from Si-LNOI have been published. Ultrahigh efficiency second harmonic generation in Si-LNOI has been proposed with a normalized conversion efficiency of over 3900% $\text{W}^{-1} \cdot \text{cm}^{-2}$ [24]. A M-Z electro-optic modulator in Si-LNOI with a $V_{\pi}L$ of 1.76 V·cm was also proposed and the modulation bandwidth simulated to exceed 350 GHz [25].

A phase modulator is a basic component of M-Z modulators and electro-optic switches. In this study, we designed, fabricated and characterized phase modulators in Si-LNOI. The optical power in the LN layer and mode areas of the Si loading strip waveguides were simulated and analyzed. Because of the large refractive index of Si, the Si loading strip waveguide had compact mode areas ($0.36 \mu\text{m}^2$). This allowed the phase modulator to have a small electrode gap and a $V_{\pi}L$ as small as 3.3 V·cm. The phase modulators in Si-LNOI were fabricated by forming Si loading strip waveguides using plasma etching. The thickness and top width of the Si loading strip waveguide were 100 nm and $0.5 \mu\text{m}$, respectively. The electrode gap was $3.7 \mu\text{m}$. The measured $V_{\pi}L$ of the phase modulator was 4.74 V·cm. This study can provide a useful reference for the fabrication of devices in Si-LNOI.

2. Design and simulation

Figure 1(a) is a schematic of the phase modulator in the Si-LNOI. From top to bottom, the structure consists of a Si loading strip waveguide, LN thin film, SiO_2 layer, and Si substrate. The LN thin film was x-cut and the transverse-electric (TE) optical mode in the waveguide transmitted along the y axis in order to take advantage of a high electro-optic coefficient ($r_{33} = 28.6 \text{ pm/V}$ at 1550 nm [26]) of the LN crystal. The phase modulator consisted of a straight waveguide and electrodes on either side of the waveguide. Owing to the limited thickness of Si, part of the light resided in the LN thin film. The refractive index of the LN crystal changed when voltage was applied to the electrodes. Then, the phase of the output light in the waveguide changed due to the optical path change. Figure 1(b) shows the cross-section of the phase modulator in the Si-LNOI. To facilitate discussion, the Si waveguide had a rectangular cross section in the simulation, with the same width at the top and bottom.

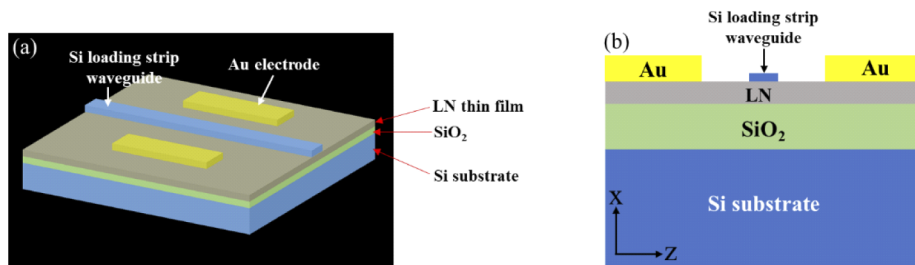


Fig. 1. (a) Schematic and (b) cross-section of phase modulator in Si-LNOI.

Half-wave voltage (V_{π}) is an important parameter of the phase modulator, and low half-wave voltage helped reduce energy consumption. In the x-cut LN thin film, the relationship between $V_{\pi}L$ and r_{33} is expressed as follows [27–28]:

$$V_{\pi}L = \frac{n_{eff}\lambda G}{n^4 r_{33} \Gamma},$$

where G is the gap between the two electrodes (electrode gap), L is the length of the electrodes, λ is the transmission wavelength, n_{eff} is the effective index, and n is the extraordinary refractive index of LN. The overlap factor Γ is expressed as follows:

$$\Gamma = \frac{G}{V} \frac{\iint E_{\text{ele}}(x, z) |E_{\text{opt}}(x, z)|^2 dx dz}{\iint |E_{\text{opt}}(x, z)|^2 dx dz},$$

where V is the voltage applied to the electrodes, E_{ele} is the electrostatic field, and E_{opt} is the optical field in the waveguide.

Large optical power in LN and compact mode areas (allowing a small electrode gap) favored a larger electrostatic field and optical field overlap, thereby reducing V_{π} . Generally, the LNOI thickness was in the range of 300 nm and 700 nm. There was a trade-off between the optical power in the LN layer and the optical mode area. In the simulation, the thickness of the LN thin film was 500 nm. The transmission wavelength was 1550 nm. The refractive indices of SiO₂ and Si were 1.46 [29] and 3.48 [23], respectively. The extraordinary refractive indexes (n_e) and ordinary refractive indexes (n_o) of LN were 2.138 and 2.211 [30], respectively. Figure 2(a) shows the relationship between TE mode optical power in LN and width of the Si loading strip waveguide. The optical power in LN decreased monotonically with increasing waveguide width.

Figure 2(b) shows the relationship between the TE mode optical mode area (defined as $\frac{(\int |E|^2 dA)^2}{\int |E|^4 dA}$ [31]) and width of the Si loading strip waveguide. The optical mode area of the Si loading strip waveguide first decreased with increasing waveguide width, and then increased with increasing waveguide width. Meanwhile, with increasing of the Si thickness, the light confinement of the waveguide became stronger due to the higher refractive index of Si, and the optical power in LN and the optical mode size of the TE mode decreased.

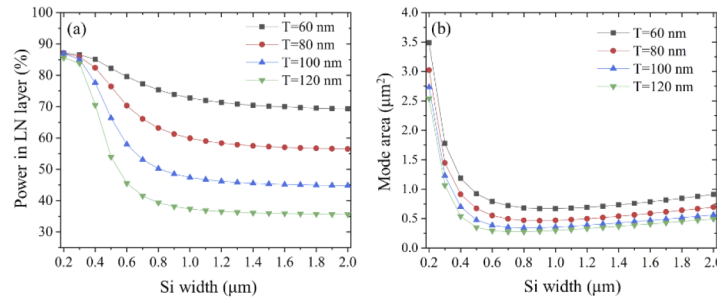


Fig. 2. Relationships of (a) TE mode optical power in LN layer and (b) TE mode area with Si loading strip waveguide width. Different color lines corresponded to different Si thicknesses (T).

Figure 3(a) shows the relationship between n_{eff} of the TE mode and width of the Si loading strip waveguide. While n_{eff} increased monotonically with increasing waveguide width, n_{eff} also increased with increasing Si thickness. A smaller electrode gap was beneficial to the enhancement of the electro-optic interaction, but it would lead to higher metal absorption loss. Here, the electrode gap was selected at the position where the metal absorption loss was 0.1 dB/cm. The electrode gap was closely related to the optical mode area. Overall, the larger optical mode area results in higher metal absorption loss when the electrode gap is constant. Meanwhile, the optical mode area was closely related to Si width. Therefore, the electrode gap depends on the width of the Si loading strip waveguide. Figure 3(b) shows the relationship between electrode gap and width of Si loading strip waveguide at different Si thicknesses. The electrode material was gold and the thickness of the electrodes was 1 μm in the simulation. First, with increasing of waveguide width, the electrode gap decreased due to the decreasing of optical mode area. Then,

the width of the Si loading strip waveguide increased further, and the electrode gap increased because of the increased optical mode area. Figure 3(c) shows the relationship between $V_{\pi}L$ and width of Si loading strip waveguide. The optical field E_{opt} was simulated by the Lumerical Mode using the full-vectorial finite difference method [31]. The electrostatic field E_{ele} was simulated by the Lumerical Charge which discretized and solved the drift-diffusion and Poisson's equations on an unstructured finite-element mesh [31]. $V_{\pi}L$ first decreased and then increased as Si width increased. At the same time, a complex relationship between $V_{\pi}L$ and Si thickness manifested. Thicker Si loading strip waveguides had a smaller optical mode area allowing for a smaller electrode gap, but the optical power in LN also decreases with increasing Si thickness. The thinner Si loading strip waveguide increased the optical power in LN, but the larger optical mode area increased the electrode gap. According to the simulation, when the thickness of Si was 100 nm and the width was 1.1 μm , $V_{\pi}L$ had a minimum value of 3.3 V·cm (the electrode gap was 2.2 μm).

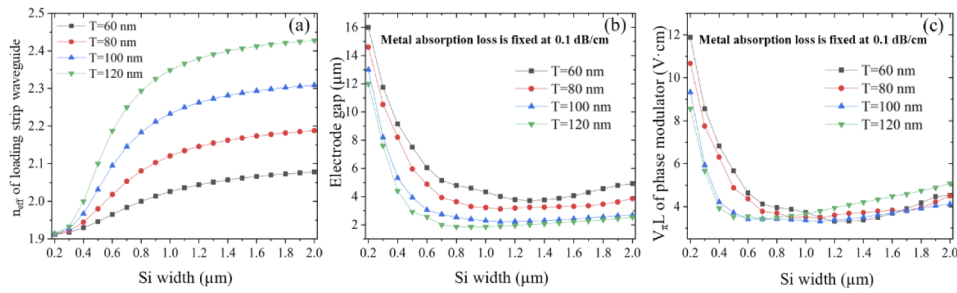


Fig. 3. Relationships of (a) n_{eff} of Si loading strip waveguide, (b) electrode gap when the metal absorption loss is fixed at 0.1 dB/cm, and (c) $V_{\pi}L$ of phase modulator with Si loading strip waveguide width. Different color lines corresponded to different Si thicknesses (T).

3. Experiments and discussion

The Si thin film was bonded on the LNOI to form a hybrid silicon and lithium niobate thin film (Si-LNOI) [22]. Figure 4(a) shows a cross-section of the Si-LNOI observed by scanning electron microscopy (SEM). The LN thin film was x-cut and the thickness of the Si thin film was 100 nm. Owing to the fabrication tolerance, the thickness of the LN thin film was 530 nm. The fabrication process of the phase modulator in the Si-LNOI was shown in Fig. 4(b). First, a layer of photoresist (HSQ) was spin-coated on the surface of the sample. Then, the Si waveguide was patterned using electron beam lithography and inductively coupled plasma (ICP) etching technology. After the Si waveguide was etched, the metal electrode was shaped using a lift-off process. The Si loading strip waveguide was parallel to the y axis of the LN. Figure 5(a) is a top view of the phase modulator via by the atomic force microscopy (AFM). The gap and length of the metal electrodes were 3.7 μm and 3 mm, respectively. Figure 5(b) is a cross-sectional view of the Si loading strip waveguide observed by AFM. The top width of the Si waveguide was 0.5 μm , the height 100 nm, and the inclination angle 35°. Figure 5(c) and (d) are the top views of the phase modulator and the Si loading strip waveguide polished facet, respectively, observed through optical microscope.

Figure 6(a) shows the simulated optical TE mode of the Si loading strip waveguide at a wavelength of 1550 nm. The optical powers in LN and Si were 56% and 31%, respectively. Figure 6(b) shows the electrostatic field distribution after a 1 V voltage was applied to the electrodes. Owing to the large refractive index of Si, the optical mode area was only 0.38 μm^2 . The small optical mode area allowed a small electrode gap, enhancing the overlap between the optical field and electrostatic field. The overlapping factor Γ of the optical field and the

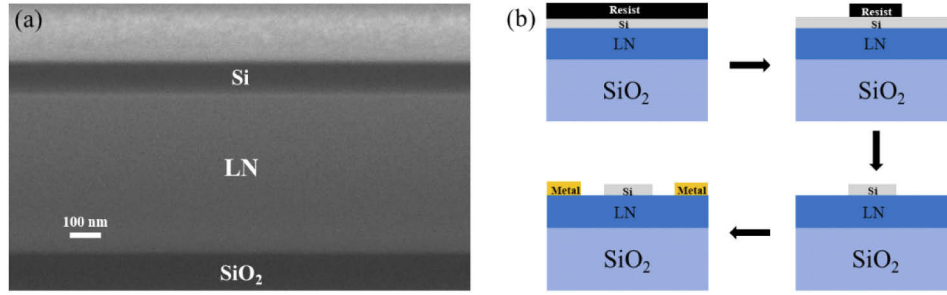


Fig. 4. (a) Cross-section of Si-LNOI observed by SEM and (b) fabrication process of the phase modulator in Si-LNOI (b).

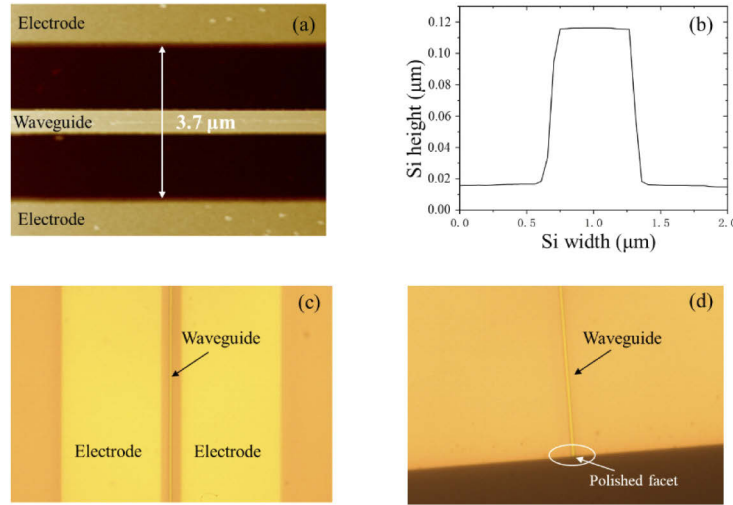


Fig. 5. (a) Top view of the phase modulator and (b) cross-sectional view of the Si loading strip waveguide observed by AFM. (c) Top view of the phase modulator and (d) the 0.5-μm-wide Si loading strip waveguide observed by optical microscope.

electrostatic field was 0.46. Figure 6(c) (black line) shows the simulated relation between electrode gap and metal absorption loss. The larger the electrode gap, the smaller the loss. The red line in Fig. 6(c) shows that the simulated $V_{\pi}L$ of the phase modulator increased with increasing electrode gap. When the electrode gap was 3.7 μm, the metal absorption loss was simulated to be 0.003 dB/cm and the simulated $V_{\pi}L$ was 4.4 V·cm; $V_{\pi}L$ was higher than 3.3 V·cm in the simulation due to the selection of a larger electrode gap (instead of 2.2 μm) to provide a degree of fabrication tolerance to avoid extra metal absorption loss. As shown in Fig. 7, the transmission loss of the waveguide was measured by the Fabry-Perot resonator method [23], which was 5 dB/cm. The transmission loss might be due to the scattering caused by the roughness of the side wall of the waveguide and by metal absorption loss.

The two polished end faces of the waveguide formed a low-finesse Fabry-Perot cavity [32]. By changing the voltage applied to the electrodes, the refractive index of LN would change, thus changing the optical path of the light and causing an output light intensity fluctuation due to interference, forming an oscillating curve. The optical path difference between any two adjacent emerging lights in the waveguide is:

$$\Delta L = 2n_{eff}L \quad (.)$$

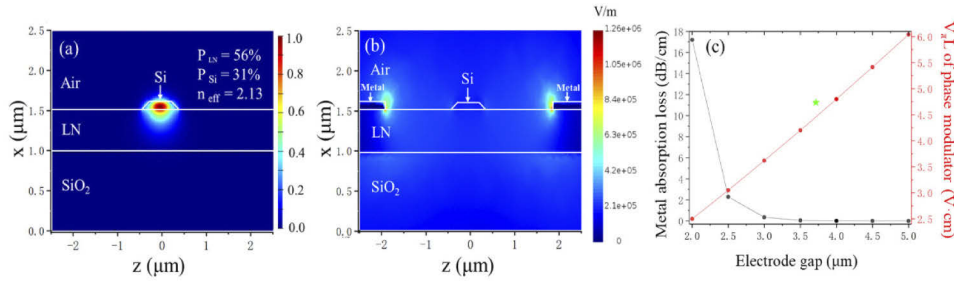


Fig. 6. (a) Optical mode profile of TE mode at 1550 nm and (b) electrostatic field after 1-V voltage was applied to the electrodes of the phase modulator. (c) Metal absorption loss (black line) and $V_{\pi}L$ (red line) of the phase modulator vary with the electrode gap.

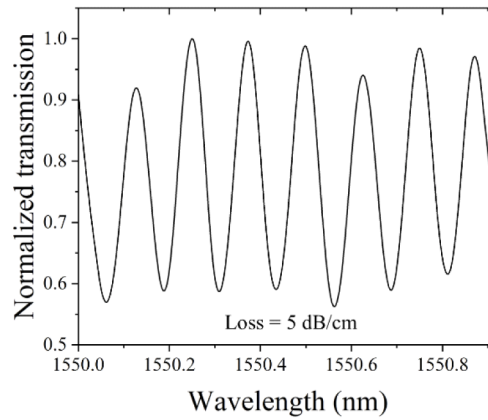


Fig. 7. Normalized transmission of the Si loading strip waveguide as a function of wavelength.

As shown in Ref. [27], when a voltage was applied to the electrodes,

$$\Delta n_{eff} = \Delta n_0 V, \quad \Delta n_0 = \frac{-k_0 r_{33} n_e^4 \Gamma}{2\beta G},$$

where β is the phase constant and k_0 is the vacuum wavenumber. Therefore, the phase difference is:

$$\delta = \frac{4\pi}{\lambda} (n_{eff} + \Delta n_0 V) L = A + BV,$$

$$A = \frac{4\pi}{\lambda} n_{eff} L,$$

$$B = \frac{4\pi}{\lambda} \Delta n_0 L.$$

The transmitted light in the waveguide is:

$$T = \frac{1}{1 + F \sin^2\left(\frac{\delta}{2}\right)},$$

$$F = \frac{4R}{(1 - R)^2},$$

where R is the reflectivity.

Using this method, V_π of the phase modulator can be obtained and the electro-optical (E-O) coefficient of LN thin film calculated. The measurement system is shown in Fig. 8(a). Linearly TE polarized light was emitted from a tunable semiconductor laser and transmitted through a polarization maintaining fiber. The light was coupled into the waveguide by a lensed fiber, and the coupling efficiency with the Si loading strip waveguide was 7 dB per facet. The output of the waveguide was collected by a 40× objective lens and detected by a Ge detector. A triangular wave signal (10-kHz) from a signal generator was amplified by a high-voltage amplifier and applied to the metal electrodes. Figure 8(b) shows a 10-kHz triangular wave sweep with an amplitude of 26 V. The black line is the triangular electric signal and the red line is the detected optical signal. The measured V_π of the phase modulator was 15.8 V and its $V_\pi L$ was 4.74 V·cm [as shown the green star in Fig. 6(c)], corresponding to 2.37 V·cm for M-Z modulator. This value was close to that of the modulator in LN thin film [33]. The E-O coefficient r_{33} of LN thin film was evaluated to be 27 pm/V, which was close to than that of the bulk material. There are two possible reasons for the E-O coefficient decrease: one is the lattice damage of LN caused by ion implantation and the other is that the optical modes in the Si loading strip waveguide changed due to the uneven thickness of Si, which would result in the simulation error in overlapping factor Γ . In addition, the wavelength range of the phase modulator was measured in the range of 1360-1630 nm. As shown in Fig. 9(a), n_{eff} of the TE mode decreases with increasing wavelength, and the optical power of the TE mode in LN increases with the increase of wavelength. As shown in Fig. 9(b), the $V_\pi L$ of the phase modulator changed from 5.5 to 4.4 V·cm in the range of 1360-1630 nm, indicating the wide working wavelength range.

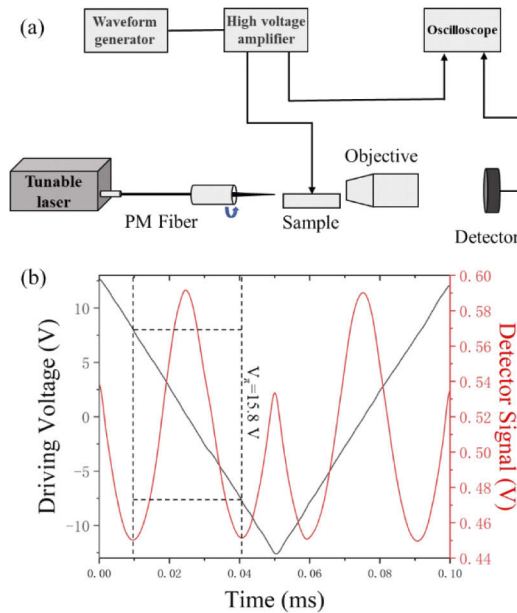


Fig. 8. (a) Measurement system of V_π . (b) 10-kHz triangular wave sweep with an amplitude of 26 V for the 3-mm-long phase modulator: black line, triangular electric signal; and red line, output optical signal.

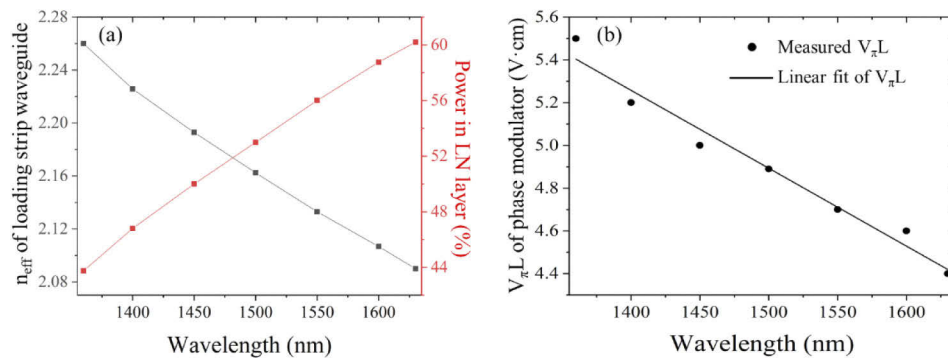


Fig. 9. (a) Relationships of n_{eff} of Si loading strip waveguide and optical power in LN layer of TE mode with wavelength. (b) Relationship of $V_{\pi}L$ of phase modulator with wavelength.

4. Conclusions

In conclusion, a phase modulator in Si-LNOI was demonstrated. Owing to the large refractive index of Si, the compact mode areas of the Si loading strip waveguide allowed a small electrode gap to enhance the overlap of the optical field and the electrostatic field. The simulated $V_{\pi}L$ of the phase modulator could be as low as 3.3 V·cm. Phase modulators with a Si loading strip waveguide were fabricated by plasma etching. The $V_{\pi}L$ of the phase modulator in Si-LNOI was 4.74 V·cm, corresponding to 2.37 V·cm for M-Z modulator, which was close to that of the modulator in LN thin film.

Funding. Natural Science Foundation of Shandong Province (ZR2020LLZ007); National Key Research and Development Program of China (2018YFB2201700, 2019YFA0705000).

Acknowledgements. The authors thank Y. F. Zhang for his kind help and discussions on device characterization.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. R. S. Weisand and T. K. Gaylord, "Lithium niobate: summary of physical properties and crystal structure," *Appl. Phys. A* **37**(4), 191–203 (1985).
2. L. Arizmendi, "Photonic applications of lithium niobate crystals," *Phys. Status Solidi A* **201**(2), 253–283 (2004).
3. R. V. Schmid and I. P. Kaminow, "Metal-diffused optical waveguides in LiNbO₃," *Appl. Phys. Lett.* **25**(8), 458–460 (1974).
4. M. L. Bortz and M. M. Fejer, "Annealed proton-exchanged LiNbO₃ waveguides," *Opt. Lett.* **16**(23), 1844–1846 (1991).
5. J. L. Jackel, C. E. Rice, and J. J. Veselka, "Proton exchange for high-index waveguides in LiNbO₃," *Appl. Phys. Lett.* **41**(7), 607–608 (1982).
6. F. Chen, "Photonic guiding structures in lithium niobate crystals produced by energetic ion beams," *J. Appl. Phys.* **106**(8), 081101 (2009).
7. M. Levy, R. M. Osgood, R. Liu, L. E. Cross, G. S. Cargill, A. Kumar, and H. Bakhru, "Fabrication of single crystal lithium niobate films by crystal ion slicing," *Appl. Phys. Lett.* **73**(16), 2293–2295 (1998).
8. G. Poberaj, H. Hu, W. Sohler, and P. Gunter, "Lithium niobate on insulator (LNOI) for micro-photonics devices," *Laser Photonics Rev.* **6**(4), 488–503 (2012).
9. A. J. Mercante, P. Yao, S. Shi, G. Schneider, J. Murakowski, and D. W. Prather, "110 GHz CMOS compatible thin film LiNbO₃ modulator on silicon," *Opt. Express* **24**(14), 15590–15595 (2016).
10. C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Loncar, "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages," *Nature* **562**(7725), 101–104 (2018).
11. M. Y. Xu, M. B. He, H. G. Zhang, J. Jian, Y. Pan, X. Y. Liu, L. F. Chen, X. Y. Meng, H. Chen, Z. H. Li, X. Xiao, S. H. Yu, S. Y. Yu, and X. L. Cai, "High-performance coherent optical modulators based on thin-film lithium niobate platform," *Nat. Commun.* **11**(1), 3911 (2020).

12. M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Loncar, "Monolithic ultra-high-Q lithium niobate microring resonator," *Optica* **4**(12), 1536–1537 (2017).
13. J. Wang, F. Bo, S. Wan, W. X. Li, F. Gao, J. J. Li, G. Q. Zhang, and J. J. Xu, "High-Q lithium niobate microdisk resonators on a chip for efficient electro-optic modulation," *Opt. Express* **23**(18), 23072–23078 (2015).
14. Z. W. Fang, N. Yao, M. Wang, J. T. Lin, J. H. Zhang, R. B. Wu, L. L. Qiao, W. Fang, T. Lu, and Y. Cheng, "Fabrication of high quality factor lithium niobate double-disk using a femtosecond laser," *Int. J. Optomechatron.* **11**(1), 47–54 (2017).
15. L. Chang, Y. F. Li, N. Volet, L. R. Wang, J. Peters, and J. E. Bowers, "Thin film wavelength converters for photonic integrated circuits," *Optica* **3**(5), 531–535 (2016).
16. C. Wang, C. Langrock, A. Marandi, M. Jankowski, M. Zhang, B. Desiatov, M. M. Fejer, and M. Loncar, "Ultrahigh-efficiency wavelength conversion in nanophotonic periodically poled lithium niobate waveguides," *Optica* **5**(11), 1438–1441 (2018).
17. Y. T. Niu, C. Lin, X. Y. Liu, Y. Chen, X. P. Hu, Y. Zhang, X. L. Cai, Y. X. Gong, Z. D. Xie, and S. N. Zhu, "Optimizing the efficiency of a periodically poled LNOI waveguide using in situ monitoring of the ferroelectric domains," *Appl. Phys. Lett.* **116**(10), 101104 (2020).
18. A. Rao, K. Abdelsalam, T. S. Aardema, A. Honardoost, G. F. Camacho-Gonzalez, and S. Fathpour, "Actively-monitored periodic-poling in thin-film lithium niobate photonic waveguides with ultrahigh nonlinear conversion efficiency of $4600\%W^{-1}cm^{-2}$," *Opt. Express* **27**(18), 25920–25930 (2019).
19. J. Zhao, M. Rusing, U. A. Javid, J. W. Ling, M. X. Li, Q. Lin, and S. Mookherjee, "Shallow-etched thin-film lithium niobate waveguides for highly-efficient second-harmonic generation," *Opt. Express* **28**(13), 19669–19682 (2020).
20. L. Chen, J. Chen, J. Nagy, and R. M. Reano, "Highly linear ring modulator from hybrid silicon and lithium niobate," *Opt. Express* **23**(10), 13255–13264 (2015).
21. P. O. Weigel, J. Zhao, K. Fang, H. Al-Rubaye, D. Trotter, D. Hood, J. Mudrick, C. Dallo, A. T. Pomerene, A. L. Starbuck, C. T. DeRose, A. L. Lentine, G. Rebeiz, and S. Mookherjee, "Bonded thin film lithium niobate modulator on a silicon photonics platform exceeding 100 GHz 3-dB electrical modulation bandwidth," *Opt. Express* **26**(18), 23728–23739 (2018).
22. H. B. Zhu, Q. Y. Li, H. P. Han, Z. Y. Li, X. Q. Zhang, H. H. Zhang, and H. Hu, "Hybrid mono-crystalline silicon and lithium niobate thin films," *Chin. Opt. Lett.* **19**(6), 060017 (2021).
23. Y. W. Wang, Z. H. Chen, L. T. Cai, Y. P. Jiang, H. B. Zhu, and H. Hu, "Amorphous silicon-lithium niobate thin film strip-loaded waveguides," *Opt. Mater. Express* **7**(11), 4018–4028 (2017).
24. B. Fang, S. L. Gao, Z. Z. Wang, S. N. Zhu, and T. Li, "Efficient second harmonic generation in silicon covered lithium niobate waveguides," *Chin. Opt. Lett.* **19**(6), 060004 (2021).
25. J. Wang, S. F. Xu, J. P. Chen, and W. W. Zou, "A heterogeneous silicon on lithium niobate modulator for ultra-compact and high-performance photonic integrated circuits," *IEEE Photonics J.* **13**(1), 1–12 (2021).
26. K. Yonekura, L. H. Jin, and K. Takizawa, "Measurement of dispersion of effective electro-optic coefficients and of non-doped congruent LiNbO₃ crystal," *J. Appl. Phys.* **47**(7), 5503–5508 (2008).
27. A. Honardoost, R. Safian, A. Rao, and S. Fathpour, "High-speed modeling of ultracompact electrooptic modulators," *J. Lightwave Technol.* **36**(24), 5893–5902 (2018).
28. N. Boynton, H. Cai, M. Gehl, S. Arterburn, C. Dallo, A. Pomerene, A. Starbuck, D. Hood, D. C. Trotter, T. Friedmann, C. T. DeRose, and A. Lentine, "A heterogeneously integrated silicon photonic/lithium niobate travelling wave electro-optic modulator," *Opt. Express* **28**(2), 1868–1884 (2020).
29. L. M. Landsberger and W. A. Tiller, "Refractive index, relaxation times and the viscoelastic model in dry-grown SiO₂ films on Si," *Appl. Phys. Lett.* **51**(18), 1416–1418 (1987).
30. D. E. Zelmon and D. L. Small, "Infrared corrected Sellmeier coefficients for congruently grown lithium niobate and 5 mol. % magnesium oxide-doped lithium niobate," *J. Opt. Soc. Am. B* **14**(12), 3319–3332 (1997).
31. Lumerical Solutions, Available online: <http://www.lumerical.com/> (accessed on 20 December 2021).
32. R. Regener and W. Sohler, "Loss in low-finesse Ti: LiNbO₃ optical waveguide resonators," *Appl. Phys. B* **36**(3), 143–147 (1985).
33. D. Zhu, L. B. Shao, M. J. Yu, R. Cheng, B. Desiatov, C. J. Xin, Y. W. Hu, J. Holzgrafe, S. Ghosh, A. Shams-Ansari, E. Puma, N. Sinclair, C. Reimer, M. Zhang, and M. Loncar, "Integrated photonics on thin-film lithium niobate," *Adv. Opt. Photonics* **13**(2), 242–352 (2021).